

P-VIDO: A Tightly-Coupled Vehicle Dynamics and Plane Priors Extension to Visual-Inertial State Estimator

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Abstract—Traditional monocular visual-inertial odometry (VIO) system cannot be applied to vehicle platforms due to unobservable degenerate motions, which result in poor initialization and low positioning accuracy. We propose the P-VIDO, a fast monocular visual-inertial odometry system with tightly coupled coplanar constraints and vehicle dynamics constraints. Plane landmarks are constructed using 3-point RANSAC methods and introduced into the optimization problem to provide coplanar constraints for point features. This significantly improves the accuracy of the odometry system in a structural environment such as an indoor parking lot. The experiments show that the system is executable in different vehicle motions with higher precision, as far as existing VIO methods are concerned. Besides, the system proposed in the paper can achieve 55fps averagely with a single thread.

Index Terms—VIO, plane, dynamics model

I. INTRODUCTION

Recent advances in visual-inertial odometry (VIO) have given birth to many high performance systems applied in autonomous driving, unmanned aerial vehicles, and augmented and virtual reality [1]–[3].

Although the fusion of IMU and camera can make the state estimation of the system more accurate and robust than that of vision-based systems [4], it is not optimized for ground vehicles because of some motions like uniform motions making some degrees of freedom in pose unobservable. Those motions can make systems fail to initialize and cause scale drift, as a result of inadequate IMU excitation [1]. Such a problem can

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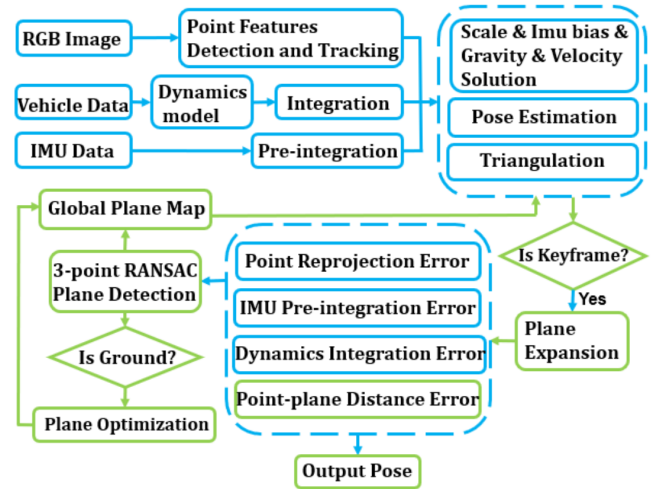


Fig. 1. The pipeline of P-VIDO. The system input is RGB images, IMU data, steering angle data and wheel speed data, which come from sensors installed on the vehicle. Different types of residual functions are introduced to build the sliding window: (a) Point Reprojection Error, (b) IMU Pre-integration Error, (c) Dynamics Integration Error, (d) Point-plane Distance Error.

be resolved by tightly coupling vehicle dynamics model into the VIO system [1].

Meanwhile, point features cannot be robustly matched and triangulated in many man-made textureless scenes, such as a parking lot, which greatly degrade the performance of a VINS-dynamics coupled system [5]. However, these scenes are usually rich in planar structures. And the combination of point and plane features can provide reliable visual information for

the VIO system [2] when navigating in environments with poor features.

Therefore, this paper proposes a tightly-coupled system (P-VIDO) that combines the vehicle dynamics and coplanar constraints in a VIO system to further improve its estimation accuracy and stability while ensuring real-time performance.

The framework of our method is shown in Figure 1. The parts marked in blue show our previous work [1], while the green parts of the figure illustrate our three main contributions:

- A VIO system tightly coupled plane constraints and vehicle dynamics constraints for vehicle states estimation in a parking lot.
- Plane information in a global plane map helps locate the current frame pose quickly. Positions of landmarks and vehicle poses can be corrected when passing through the same plane.
- Point-plane distance errors can automatically introduce plane constraints during sliding-window optimization with little extra computational cost.

II. RELATED WORK

In this section, previous work are briefly reviewed from two aspects: VIO for ground vehicles and SLAM with coplanar constraints.

A. VIO For Ground Vehicles

In recent years, several low-cost pose estimation methods have been devised for ground vehicles and robots by introducing wheel odometer measurements to VIO systems. Specifically, [6] designed to add wheel odometer measurements and bring the assumption of planar motion in state estimation, which showed the overall performance improvement in indoor scenes. Likewise, [7] proposed to combine the measurements from IMU and wheel odometer with vision-based observations, which help the system to integrate the current pose and correct the recovered visual scale. In addition, in order to introduce the wheel odometer measurements more accurately and improve the overall accuracy of the system, the relative external parameters among different sensors are also estimated online as optimization variables [8]. Instead of assuming that the vehicle moves on a ground plane, [9] suggested to accomplish all frame poses reckoning and the ground manifold parameters estimation jointly at the same time. The local motion manifold was modeled by second-order polynomial equations. While above work is comprehensive for most ground vehicles, the cornering properties of pneumatic tires require to be taken into additional consideration, especially when the vehicle is turning. Our previous work VIDO [1] employed the vehicle dynamics model into VIO to build an extra odometer constraints, which help the system initialize robustly.

B. SLAM with Coplanar Constraints

Assuming the existence of a dominant global plane in the scene, GPO [10] proposed to estimate planes and poses in the way of global optimization, and then it recovered map

points directly from planes. Based on Superpixel, TT-SLAM [11] introduced planar template-based trackers for tracking planes in consecutive image frames, which can address data association and keyframe selection efficiently, compared to RANSAC-based methods. However, texture-less scenes limit the use of superpixels.

PVIO [2] proposed a novel method to introduce plane information to localize single frame fast and robust. Planes involved in pose optimization are extracted from point landmarks by 3-point RANSAC method. By including planar priors constraints in single-frame bundle adjustment, the system achieve a smooth and fast front-end process with no global map to be maintained anymore. Inspired by this work, we introduce coplanar constraints in a texture-less but plane information-rich environment, like a parking lot, to improve the performance of VIDO system. Planar features are combined with single-point features using a coherent initialization process, which allows point features to grow into the plane feature when appropriate. Furthermore, we model the ground plane to optimize the normal vector of the continuous plane directly and quickly.

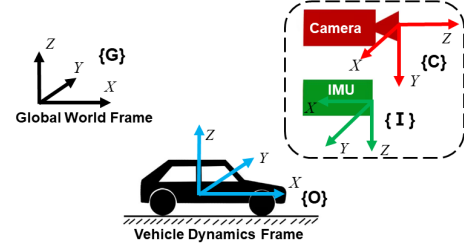


Fig. 2. Conceptual representation of a ground vehicle and sensors in the vehicle. The global reference frame $\{G\}$, vehicle dynamics frame $\{O\}$, IMU frame $\{I\}$ and camera frame $\{C\}$ are represented in different colors. The vehicle moves approximately on a plane with which wheels of the vehicle are always in contact.

III. P-VIDO FRAMEWORK

A. Notations

In this work, we presume a ground vehicle positioning relative to a global world reference frame, $\{G\}$. $\{O\}$ is used to indicate vehicle reference frames, the local coordinate system of the wheel odometer. Its x-axis points forward and z-axis points up (see Fig. 2 for details). The reference frame of the monocular camera and IMU, is denoted by $\{C\}$ and $\{I\}$ respectively. We use ${}^G\mathbf{p}_i$ and ${}^G\mathbf{q}_i$ to represent the i^{th} camera frame pose. Rigid extrinsic of camera and IMU relative to vehicle dynamics coordinate ${}^O\mathbf{R}$ (or ${}^O\mathbf{R}_0$) and ${}^I\mathbf{R}$ (or ${}^I\mathbf{R}_0$) can be calibrated beforehand.

B. Visual-Inertial-Dynamics Odometry

1) *The 2-DOF Vehicle Dynamics Model:* We adopt the linear vehicle dynamics model [1] to represent the motion of the vehicle on the plane, as described in Figure 3.

When vehicle rotating at a constant angular velocity, its steady response under front wheel angle step inputs is uniform

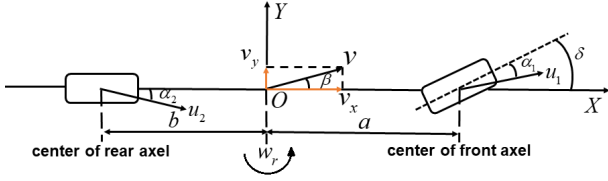


Fig. 3. The 2-DOF dynamics model. α_1 and α_2 denote the sideslip angle of the front and rear axles, whose speed at the midpoint are u_1 and u_2 . ω_r and δ represent the yaw velocity and front wheel angle respectively. The centroid velocity can be decomposed into v_x and v_y .

circular motion. At this time, v_y and ω_r are constant. Then the angle β and the ratio of the steady-state yaw rate to the front wheel angle $\left(\frac{\omega_r}{\delta}\right)_s$ are:

$$\beta = \frac{(ak_1 - \frac{a^2k_1 + b^2k_2}{(1+Kv^2)(a+b)})\delta}{ak_1 - bk_2} \quad (1)$$

$$\left(\frac{\omega_r}{\delta}\right)_s = \frac{v_x/(a+b)}{1 + Kv_x^2} = \frac{v \cos \beta / (a+b)}{1 + K(v \cos \beta)^2}$$

where $K = \frac{m}{(a+b)^2} \left(\frac{a}{k_2} - \frac{b}{k_1}\right)$, and i is the ratio of front wheel angle to steering wheel angle. Vehicle dynamics model measurements are given by:

$$\hat{\mathbf{v}} = [v \cos \beta \quad v \sin \beta \quad 0]^T \quad (2)$$

$$\hat{\boldsymbol{\omega}} = [0 \quad 0 \quad \omega_r]^T$$

2) *Vehicle Dynamics Integration*: When the system states at time k are known, combining measurements obtained from the vehicle dynamics model, the vehicle pose at time $k+1$ can be deduced as:

$$\begin{aligned} {}^G\mathbf{p}_{k+1} &= {}^G\mathbf{p}_k + \int_{t_k}^{t_{k+1}} [{}^G\mathbf{R}_t(\hat{\mathbf{v}}_t - \mathbf{n}_v)] dt \\ {}^G\mathbf{q}_{k+1} &= {}^G\mathbf{q}_k \otimes \int_{t_k}^{t_{k+1}} [{}^G\mathbf{q}_t[(\hat{\boldsymbol{\omega}}_t - \mathbf{n}_\omega)]] dt \end{aligned} \quad (3)$$

where $\mathbf{n}_v \sim N(0, \sigma_v^2)$ and $\mathbf{n}_\omega \sim N(0, \sigma_\omega^2)$, modeled as gaussian white noise. $[\boldsymbol{\omega}]$ satisfies:

$$[\boldsymbol{\omega}] = \frac{1}{2} \begin{bmatrix} 0 & -\boldsymbol{\omega}^T \\ \boldsymbol{\omega} & \boldsymbol{\omega}^\wedge \end{bmatrix} \quad (4)$$

In time interval $[t_k, t_{k+1}]$, the integration term of pose are:

$$\begin{aligned} {}^{O_k}_{O_{k+1}}\boldsymbol{\eta} &= \int_{t_k}^{t_{k+1}} [\mathbf{R}_t(\hat{\mathbf{v}}_t - \mathbf{n}_v)] dt \\ {}^{O_k}_{O_{k+1}}\boldsymbol{\theta} &= \int_{t_k}^{t_{k+1}} [\mathbf{q}_t[(\hat{\boldsymbol{\omega}}_t - \mathbf{n}_\omega)]] dt \end{aligned} \quad (5)$$

Then we can calculate the jacobians by deriving discrete-time error-state kinematics in time interval $\Delta t = t_{k+1} - t_k$:

$$\begin{bmatrix} \delta \boldsymbol{\eta}_{k+1} \\ \delta \boldsymbol{\theta}_{k+1} \end{bmatrix} = \mathbf{A}_k(t) \begin{bmatrix} \delta \boldsymbol{\eta}_k \\ \delta \boldsymbol{\theta}_k \end{bmatrix} + \mathbf{B}_k(t) \begin{bmatrix} \mathbf{n}_v \\ \mathbf{n}_\omega \end{bmatrix} \quad (6)$$

where

$$\mathbf{A}_k(t) = \begin{bmatrix} \mathbf{I} & -\mathbf{R}_k(\hat{\mathbf{v}}_k)^\wedge \Delta t \\ 0 & \mathbf{I} - (\hat{\boldsymbol{\omega}}_k)^\wedge \Delta t \end{bmatrix}, \quad \mathbf{B}_k(t) = \begin{bmatrix} -\mathbf{R}_k \Delta t & 0 \\ 0 & -\Delta t \end{bmatrix} \quad (7)$$

After dynamics integration, we utilize the integration result to align vision and IMU outputs separately for initialization, which is the same as our previous work [1].

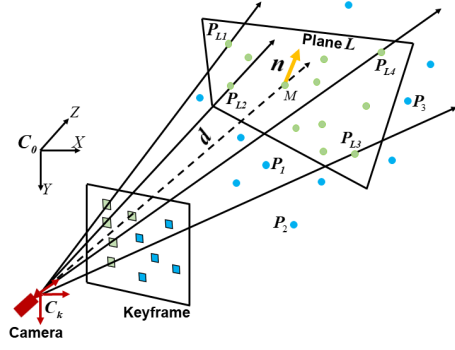


Fig. 4. Co-planar points fit plane parameters. The green points are on the same plane landmark, while the blue points are not on the plane. Distance between plane L and current camera center C_k is $d = |MC_k|$.

C. Coplanar constraint in Visual-Inertial-Dynamics Odometry

1) *Plane-aided Current Pose Recovery*: In this paper, key-points in the current image frame k are tracked using a pyramidal tracking algorithm [12] and represented as $\mathbf{m}_k$. With the pinhole camera model where camera intrinsic K is fixed, the projection of triangulated point landmark $\mathbf{P}_h$ is:

$$\begin{aligned} \hat{\mathbf{m}}_{kh} &= K \cdot {}^G_C \mathbf{R}_k (\mathbf{P}_h - {}^G_C \mathbf{p}_k) \\ \mathbf{P}_h &= \frac{1}{\lambda} [u \quad v \quad 1]^T \end{aligned} \quad (8)$$

where $[u \quad v \quad 1]^T$ is the homogeneous coordinate of $\mathbf{P}_h$ in the image frame k and λ is its inverse depth.

A plane L is parameterized with its normal $\mathbf{n}_L$ and distance d_L to the origin of camera, as described in Figure 4. Landmark point sets are divided into two categories, the point set $\{\mathbf{P}_{Lh} : h = 1 \dots N_L\}$ belonging to the plane L and the point set $\{\mathbf{P}_h : h = 1 \dots M\}$ not lying on any plane. Points on plane L should satisfy:

$$\mathbf{n}_L^T \cdot \mathbf{P}_{Lh} - d_L = 0 \quad (9)$$

We cast a ray from current camera center C_k , and find the intersection depth $\tilde{\lambda}_h$ of point h with planes in the global plane map, as described in Figure 4. As a result, the inverse depth of points are obtained based on the assumption that they belong to the plane:

$$\tilde{\lambda}_h = \frac{\mathbf{n}_L^T \cdot {}^G_C \mathbf{R}_k [u \quad v \quad 1]^T}{d_L - \mathbf{n}_L^T \cdot {}^G_C \mathbf{p}_k} \quad (10)$$

By combining Equation 8 and Equation 10, the point reprojected with $\tilde{\lambda}_h$ is represented as:

$$\tilde{\mathbf{m}}_{kh} = K \cdot {}^G_C \mathbf{R}_k \left(\frac{1}{\tilde{\lambda}_h} [u \quad v \quad 1] - {}^G_C \mathbf{p}_k \right) \quad (11)$$

Plane information managed in the global plane map helps locate the camera pose quickly when new frame coming. Since triangulation errors easily cause large depth estimation errors for point landmarks, we adopt point-plane distance method to add coplanar constraints into the current pose solution. We can

re-cast the landmark point $\mathbf{P}_h$ onto plane L on the basis of Equation 10. Reprojection residuals is computed as:

$$\begin{aligned} E_{point} &= \sum_{h=1}^M \|m_{kh} - \hat{m}_{kh}\|_{Q_1}^2 \\ E_{plane} &= \sum_{L=1}^n \sum_{h=1}^{N_L} \|m_{k,Lh} - \tilde{m}_{k,Lh}\|_{Q_2}^2 \end{aligned} \quad (12)$$

where n is plane numbers in the global plane map.

We use E_{imu} to represent the IMU propagation error with corresponding covariance Q_{imu} . Therefore, current frame pose can be obtained by solving the following local bundle adjustment:

$$\arg \min_{\substack{O_k, \mathbf{p}_k, \mathbf{q}_k}} \left\{ \|E_{imu}\|_{Q_{imu}}^2 + \sum_{h=1}^M \|E_{point}\|_{Q_1}^2 + \sum_{L=1}^n \sum_{h=1}^{N_L} \|E_{plane}\|_{Q_2}^2 \right\} \quad (13)$$

2) *Plane Extraction and Expansion*: For plane extraction, the subset containing three landmark points is used to compute the plane model, and coplanar points selected from other point landmarks will be added to the subset. The minimum iteration number for extracting a plane with a confidence level η that the plane is outlier-free is given by:

$$M_{Iteration} = \frac{\log(1 - \eta)}{\log(1 - r^3)} \quad (14)$$

where the true inlier ratio for extracting the L th plane is $r = 1/(N + L - 1)$ for $L = 1, \dots, N$.

Any new frame regarded as a keyframe will take part in optimization. We select new triangulated point features in the current keyframe, which match the plane model existing in the global plane map, and then we merge the points into the corresponding plane. In order to reduce the depth errors from point features triangulation, we compare the cast reprojection error with the re-cast reprojection error to decide whether the new point feature belongs to the plane.

$$\mathbf{P}_h \in \{\mathbf{P}_{Lh} : h = 1 \dots N_L\} \quad \text{if } E_{plane} \leq \alpha E_{point} \quad (15)$$

where α is set to 1.3 for consistency check.

Generally, the camera is fixed on the vehicle and parallel to the ground. Ground normal vector can be denoted as $\mathbf{n}_g = [0 \ 1 \ 0]^T$ in camera coordinate system. When a plane is judged as the ground, its normal will be represented as $\mathbf{n}_g$, and the only parameter d_g is continuously optimized as the number of coplanar points increases.

3) *Sliding-Window Optimization*: Bundle adjustment in the sliding window of the P-VIDO consists of five parts, as shown in Figure 5. Entire state variables of the proposed P-VIDO system include depths of landmark points, plane parameters, and the states related to keyframe poses, which will be optimized when single-frame or all keyframes bundle adjustment. The states at time t are defined as:

$$\begin{aligned} \mathbf{x}_t &= \{\mathbf{x}_i, \lambda_k, {}^G\pi_L\}_{i \in \mathcal{I}_t, k \in \mathcal{K}_t, L \in \pi_t} \\ \mathbf{x}_t &= [{}^G\mathbf{p}_i, {}^G\mathbf{q}_i, {}^G\mathbf{v}_i, \mathbf{b}_{a_i}, \mathbf{b}_{g_i}] \end{aligned} \quad (16)$$

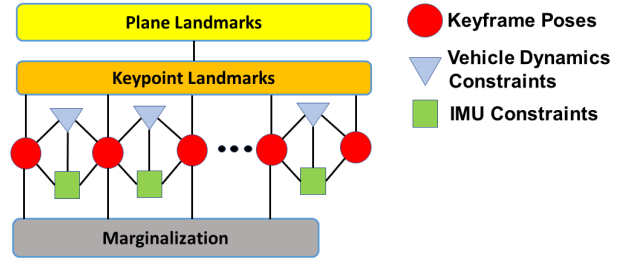


Fig. 5. Factors that consist of bundle adjustment in P-VIDO. Keyframe poses are optimized by coplanar constraints from plane landmarks, reprojection constraints from keypoint landmarks, dynamics integration constraints, pre-integration constraints from IMU and prior marginalization constraints.

The sets π_t include plane states at time t . $\mathbf{x}_i$ is composed of i^{th} IMU position ${}^G\mathbf{p}_i$, orientation ${}^G\mathbf{q}_i$, velocity ${}^G\mathbf{v}_i$, accelerometer bias $\mathbf{b}_{a_i}$ and gyroscope bias $\mathbf{b}_{g_i}$.

Given body poses of two adjacent frames, O_k and O_{k+1} , the integration residual of vehicle dynamics model is represented as:

$$E_{Dynamics} = \begin{bmatrix} O_k \\ O_{k+1} \end{bmatrix} \begin{bmatrix} \boldsymbol{\eta} \\ \boldsymbol{\theta} \end{bmatrix} = \begin{bmatrix} {}^G R({}^{O_{k+1}}_{{}^{O_k}} \mathbf{p} - {}^G \mathbf{p}) - {}^{O_k}_{O_{k+1}} \hat{\boldsymbol{\eta}} \\ 2({}^{O_k}_{O_{k+1}} \boldsymbol{\theta}^{-1} \otimes {}^G \mathbf{q}^{-1} \otimes {}^G \mathbf{q}) \end{bmatrix} \quad (17)$$

We add the vehicle dynamics measurements residual and point-plane distance residual to the VIO system. All the state variables are solved by minimizing the sum of cost terms from all the measurement errors:

$$\begin{aligned} \mathbf{x} = \arg \min_{\mathbf{x}} \{ & \sum_{h=1}^M \|E_{point}\|_{Q_1}^2 + \sum_{L=1}^n \sum_{h=1}^{N_L} \|E_{plane}\|_{Q_2}^2 \\ & + \sum \|E_{Dynamics}\|_{Q_{Dynamics}}^2 + \sum \|E_{imu}\|_{Q_{imu}}^2 + \|\mathbf{r}_p - \mathbf{H}_p \mathbf{x}\|^2 \} \end{aligned} \quad (18)$$

where $\mathbf{r}_p$ and $\mathbf{H}_p$ are the prior residual and Jacobian of marginalization operator, respectively. The Ceres Solver is used to solve this nonlinear problem.

IV. EXPERIMENTAL RESULTS

Our previous work VIDO [1] is a real-time system, which can initialize robustly and can be applied in any vehicle motions. We add plane information and plane management strategy into VIDO [1] framework to improve its performance in the structural environments, such as parking lots. The datasets are obtained with TiEV [3] platform and experimental scenes are aimed at underground garages in Tongji University. The platform equips ZED2¹ for collection of monocular images and IMU data, as shown in Figure 6. The trajectory accuracy and runtime are analyzed.

Since there is no satellite positioning signal indoors and lidar data is reliable in static scenarios, we consider using SC-LeGO-LOAM [13] to get the ground truth. All the experiments of parking lots were performed on a laptop with an Intel Core i5-6200U CPU, at 2.30 GHz, with 8 GB memory. The

¹<https://www.stereolabs.com/zed-2/>

estimated trajectories errors of different parking lots datasets are evaluated by evo [14].

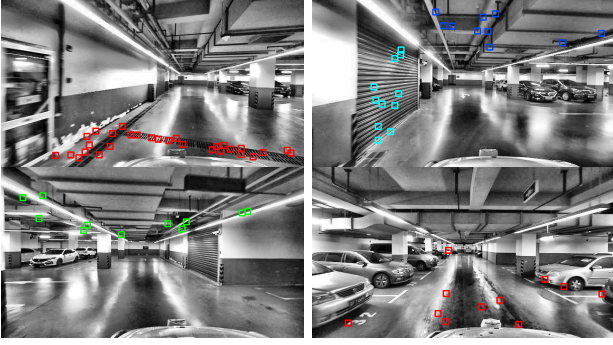


Fig. 6. Images captured in experimental scenes. Points on different planes are represented with different colors.

We compare the results with VIDO, which is the baseline of our proposed extension. we tested VIDO (without plane) and P-VIDO (with plane) to evaluate the effectiveness of vehicle dynamics extension and plane constraints separately. Loop-closure and relocation module are not included in either of them.

A. Tracking Accuracy

In this paper, we collect seven sequences for testing. The estimated trajectories from PL01 (Parking Lot1) dataset to PL07 (Parking Lot7) dataset are shown in Figure 7. The APE compared P-VIDO with VIDO in sequence PL07 is shown in Figure 9. In Table I, we summarize the RMSE of ATE among two compared systems and PVIO [2] in all sequences. It shows the proposed system achieves highly competitive accuracy.

There are two reasons. First, degenerate vehicle motions make PVIO system fail to estimate trajectory, and the initial vision scale estimated by IMU with inadequate excitation is much bigger than the real vision scale. Second, when vehicle tire parameters are changed, planar information assist the VIDO system to initialize and optimize more accurate and robust, since depths of coplanar point landmarks are further optimized with the help of planar constraints.

TABLE I
RMSE[M] IN DIFFERENT DATASETS.

Sequence	P-VIDO	VIDO	PVIO	Length(m)
PL01	0.700	0.846	>10	160.214
PL02	0.901	1.016	>10	142.348
PL03	1.467	1.644	>10	264.786
PL04	0.921	1.374	>10	276.573
PL05	1.231	1.478	>10	409.044
PL06	1.156	1.368	>10	303.680
PL07	0.861	1.374	>10	297.457

B. Efficiency

From Figure 8, we can observe that our system detects and expands planes efficiently, and the system runtime is under 22ms overall. While plane information added in the sliding

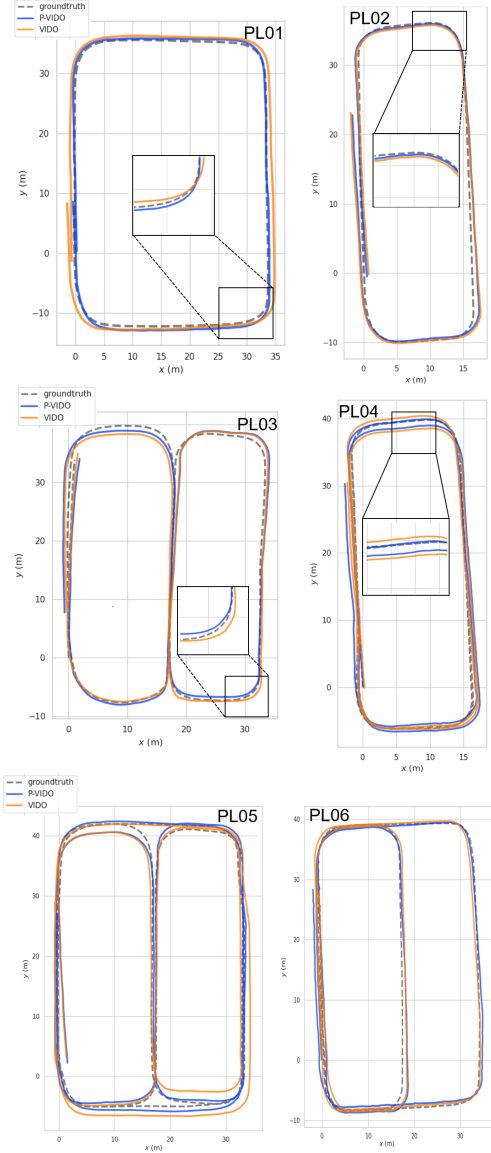


Fig. 7. The estimated trajectories from PL01 to PL06.

window optimization, the problem complexity increases not much. On the one hand, we use point-plane distance errors during sliding-window optimization process to introduce the information that the landmark points lie on the plane. On the other hand, plane information in the global plane map helps to locate the current frame pose quickly and accelerate the process of points triangulation. Besides, when passing through the same plane, the features points and camera poses can be corrected, without saving the image descriptors to detect loop-closure.

V. CONCLUSIONS AND FUTURE WORK

We have proposed P-VIDO, a fast monocular visual-inertial odometry system with tightly coupled coplanar constraints and vehicle dynamics constraints. The measurements from sensors installed on the vehicle are brought in the system

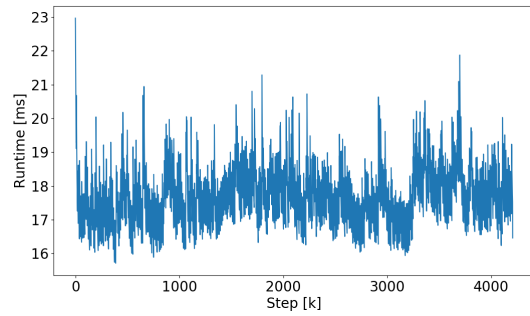


Fig. 8. Real time performance of P-VIDO in sequence PL07.

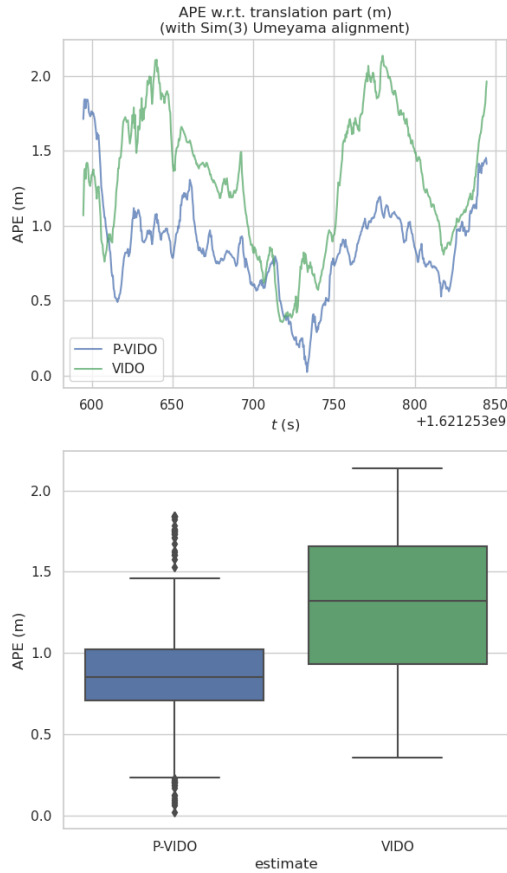


Fig. 9. The APE compared P-VIDO with VIDO in sequence PL07.

in the form of a vehicle dynamics model and then build an extra integration factor to help the system initialize. 3-point RANSAC is introduced to extract and expand planes. Plane information managed in a global plane map helps locate the camera pose quickly.

Experiments dominate that the utilization of vehicle dynamics model and planar information brings great improvement to the VIO system in both accuracy and robustness. Then we plan to add loop-closure detection into P-VIDO to further improve the system accuracy whilst ensuring high efficiency.

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